Teaching, Learning & Development Plan

M.C.A. (3rd Semester)

CS8034 Project-II

Course Objective: To imbibe essentials for requirement analysis, system modelling skill, integrating business logic.

Course Outcomes: Upon completion of the course, the student shall be able to

CO1:	Comprehend the need of the system by technical writing skill and preparation of System Requirement Specification.
CO2:	Determine the functional and non-functional requirements which reflect smartness and intelligence of the system.
CO3:	Model the functional requirements into system flow using diagrams.
CO4:	Apply data modelling process to construct the data and knowledge repository for the system.
CO5:	Implement the solution with latest software development environment along with code convention, style guide and user interface design standards.
CO6:	Follows practices that enable productive team work and effective communication with range of audiences through appropriate aids.

Programme Outcomes: The student will have

PO1:	Computational Knowledge: Apply knowledge of computing fundamentals, computing specialisation, mathematics, and domain knowledge appropriate for the computing specialisation to the abstraction and conceptualisation of computing models from defined problems and requirements.
PO2:	Problem Analysis: Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
PO3:	Design /Development of Solutions: Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO4:	Conduct Investigations of Complex Computing Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of t h e information to provide valid conclusions.
PO5:	Modern Tool Usage: Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations.
PO6:	Professional Ethics: Understand and commit to professional ethics and cyber regulations, responsibilities, and norms of professional computing practices.

PO7:	Life-long Learning: Recognise the need, and have the ability, to engage in independent learning for continual development as a computing professional.
PO8:	Project management and finance: Demonstrate knowledge and understanding of the computing and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO9:	Communication Efficacy: Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
PO10:	Societal and Environmental Concern: Understand and assess societal, environmental, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practices.
PO11:	Individual and Team Work: Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments.
PO12:	Innovation and Entrepreneurship: Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.

Semester Skillset

- Thinking skill - Critical
- Listening Skill – Evaluative
- Writing Skill - Expository

Programme Outcomes and Course Outcomes mapping

Programme Outcomes	**Course Outcomes**					
	CO1	CO2	CO3	CO4	CO5	CO6
PO1	2	2	3	3	3	0
PO2	3	3	3	2	2	0
PO3	2	2	3	3	3	0
PO4	1	1	1	1	0	0
PO5	1	1	3	3	3	0
PO6	0	0	0	0	2	1
PO7	1	1	1	1	2	0
PO8	2	2	2	1	2	3
PO9	3	2	3	1	1	3
PO10	0	0	0	0	0	1
PO11	1	1	2	1	2	3
PO12	0	1	0	1	1	1

Project Guidelines

- The project extension activity should be designed to ensure it aligns with real-world usability.
- The project may be developed using concepts, tools, and/or technologies covered in courses completed by the student in earlier semesters, or from any other course beyond the curriculum.
- Students must continue with the same team and the same project definition as in their second semester. In case of any changes, a written application must be submitted. The decision of the project committee will be final.
- Project work shall be carried out in parallel with other course-related teaching and learning activities. Assessment of the project work will also be conducted alongside assessments of other courses.
- Data processing and CRUD operations must be implemented; however, they shall not be considered in the evaluation of the system's smartness and intelligence.
- The project team must refine the requirements analyzed in the previous semester, finalize the project requirements along with the chosen tools and technologies, and submit the finalized document duly signed by both the students and the project guide to the project committee by the end of the second week of the semester.

Project Hours

- Each student is required to dedicate a minimum of approximately 100 hours to project work.

Support and guidance team for Project

- Support and Guidance Team is created so that student can receive additional support for respective project work.

Support and Guidance team	
System Analysis and Design	BAD, JBU, RBP
Database Design	ADP, ARP, KMM
System Testing	PHP, GCP, SBT
Web based Development	MMS, RBP, GCP, RPC
Mobile based Development	ARP, PHP
Content Management over Web Application	PBG, MMS, SBT
Data Analytics	BAD, ADP, PBG
Text Processing	MMS, PBG
Image Processing	JBU, ADP
IoT based Development	MMS, ADP
Project Development Support	Respective Guide

Project Progress and Evaluation

- The project will be evaluated on a continuous basis using the following parameters:
  - Adherence to project timelines
  - Usability of the project
  - Sufficiency and depth of project work
  - Effectiveness of problem solution implementation
  - System performance, user interface, and functionality testing
  - Visualization and analytical reports
  - Quality of project documentation
  - Effectiveness of project presentations

- Each team will be allocated at least thirty minutes for the project presentation during assessment A3, followed by approximately ten minutes for the viva voce.

- To assess the progress of the project work, the following parameters must be considered:

Assessment Code	Assessment Type	Occurrence	Each Of Marks	Weightage in CIE of marks (40)	Evaluated By	Tentative Date/Week	Skill set Mapped
A1	Project Proposal Report	1	10	08X01=08	Guide	27/07/2026	Thinking, Creativity, Technical Writing Skill
A2	Document Report	2	30	06x02=12	Panel Members appointed by Project committee	17/08/2026 21/09/2026	Technical Writing Skill
A3	Project Presentation	2	50	10x2=20	Panel Members appointed by Project committee	22/08/2026 26/09/2026	Technical Skill, Professional Communication
A4	External Examination	1	60	60	External Examiners	Will be declared later	Technical Writing Skill, Technical Presentation Skill, Professional Communication

A1: Project Proposal Report

Project Proposal Report should have minimum following things: Refer Annexure – I for that.

- System Background/Introduction
- System Definition
- System Need/Objectives
- System Features / functionalities including reports
- System Scope and Limitation
- System Development Approach (Tools and Technologies)

Evaluation will be done based on following parameters:

Parameter	Weightage %
Features identification and its clarity	50
Identification of Visual Analytics Reports	30
Timeliness	10
Work Sufficiency	10

A2: Document Report

- There shall be two document report submissions during the semester.
- Students must prepare a technical document, have it verified by their guide, and submit a hard copy to the Project Coordinator.
- Document Report 1 will be a partial document, while Document Report 2 will be the final version of the document.

- The marks weightage for documentation is as follows:

Evaluation parameters for Document – 1

Parameter: Document – 1	Weightage %	Topics for submission from project report
Document Flow and Formatting	20	Up to Activity 5
System Design and Implementation Approach	40	
Data Model Description	20	
Student Reporting	10	
Timeliness	10	

Evaluation parameters for Document – 2

Parameter: Document – 2	Weightage %	Topics for submission from project report
System Implementation Description	30	Complete Document as per Project Document Report Format which may include modified version of Document – 1 with proper sequence of flow.
Result Analysis and its observation	30	
Data Representation dashboard and Reports	10	
Incorporation of suggestions given in document report 1	10	
Student Reporting	10	
Timeliness	10	

Project Activity Plan:

The following are the activities to be carried out to complete project work for the defined objectives.

SN	Activity	CO Mapped	Skill Set Mapped	Execution Week of the Semester	Responsible Person	Assessment
1.	Project Course Awareness and objective of the semester	CO1		1st	Project Committee	
2.	Project Requirement Finalization, Tools and technology update	CO1	Learning, Creative and Writing	2nd	Team	A1
3.	Project Planning with Timeline preparation	CO1	Learning and Writing	3rd	Team	
4.	Proposed Software Requirement Specification • Introduction • Proposed System Primary and Secondary Objectives, Scope and Definition • System Features	CO1, CO2, CO7	Communication Skill, Writing Skill, Technical Skill& Analysis Skill	4th to 5th	Team	

	implemented in Previous semester • System features development approach including proposed algorithm/techniques • Updated User Stories Summary • Number of functional and non-functional requirements					
5.	Data Model Specification • Description about used data set / data dictionary • Pre-processing steps performs on data set • System Diagrams to present system flow	CO1, CO2, CO3, CO7	Communication Skill, Writing Skill, Technical Skill & Analysis Skill, Collaborative Skill	4th to 5th	Team	
6.	Implementation of functionalities • GUI Design technical justification • User interface screenshots and screen image with caption. • Data from: Data store, End User, Remote Server or others. • Data To: Data	CO1, CO2, CO3, CO7	Communication Skill, Writing Skill, Technical Skill & Analysis Skill	4th to 5th	Team	

	store, End User, Remote Server or others. • Critical Validations: Validation code.					
7.	The team needs to document the codes specification with following details: • Coding standard followed for development • Critical code for functionality with justification • Guidelines for availing the usage of API, plug-ins, web services, libraries for enriching the features of the system • Customizing the add-ons to assist proposed system functionalities	CO1, CO2, CO3, CO4	Creativity and Technical Writing Skill	7th	Team	
8.	Project progression status with timeline report submission	CO1	Learning and Writing	8th	Guide	
9.	Implementation of rich visual analytics reports and Dashboard with following details: • KPI based Dashboard designing and development. Refer: https://www.sisense.com/kpi-library/ https://www.scoro.com/blog/best-kpi-dashboard-software-	CO1, CO2, CO3, CO4	Thinking and Technical Writing Skill	8th	Team	A3

	tools-reviewed/ • Dashboard View [Image with caption] ○ View-1 Name ○ Description ○ Input data ○ Process representing business logic ○ Output Data ○ Usages of Data retrieve • Screens shots of visual reports with description, Input, process, output and usage.					
10.	Result Analysis specification • Structural testing of inbuilt unit testing feature of the platform utilized for implementation. • Proper verification and validation must be tested and accordingly test cases must be prepared. • UI Automation testing with tool and test cases • Performance testing of system with tool and test cases • Result observation of design model / algorithm with proper description.	CO1, CO2, CO3, CO4	Communication Skill, Writing Skill, Technical Skill & Analysis Skill	10th	Team	A3, A4
11.	Final Spiral bound soft and hard copy	All COs	Learning and Writing	13th	Student	A4

	submission by team. • The team needs to complete the verification of project document and implementation from guide. • Collection of Institute certificate and attached it with document. • The hard bound contains full documentation.					

Project Document Report Format: Final Project Document Report must cover at least following content but should not be limited to it:

- Title Page
- Project Certificate from a guide
- Declaration Certificate
- Work Sufficiency Certificate
- Acknowledgment
- Table of content with page number
- List of Tables and Figures
- Content of Project Document Report
- References
- Glossary

*Note: Enhancement is preferred in the above specification based on the nature of the project.

Guideline for Report Formatting:

- Use A4 size page with 1" margin all sides.
- The header should include Project title and the footer should contain page number and enrolment numbers.
- Chapter name should be of Cambria font, 20 points, Bold.
- The main heading should be of Cambria font, 16 points, Bold.
- The subheading should be of Cambria font, 12 points, Bold.
- Subheading of subheading should be of Cambria font, 12 points, Bold, Italic.
- The paragraph should be of Cambria font, 12 points.
- Line spacing - 1.5 lines, before - 0, after - 0.
- No chapter number for references.
- Before chapter 1, give the page number in roman letter (Acknowledgements, and Table of Contents/Index with page numbering, List of Tables, Figures, Schemes, and Summary/abstract of the project work).
- Each project document must have a front page according to Annexure-II.
- Each project document must have a declaration certificate page according to Annexure-III.
- Each project document must have a work sufficiency certificate signed by guide according to Annexure-IV.

A3: Presentation

- There will be two presentations, each evaluated by a panel of examiners.
- Presentations must be prepared in accordance with established guidelines for effective presentations.
- No make-up work will be accepted for missed or failed presentations.
- Each presentation must be reviewed and approved by the guide prior to the scheduled presentation date.
- Presentation 1 must cover at least the following aspects:
    - Project title, team member names, enrolment numbers, guide name
    - System process flow
    - System Study
    - Proposed System Definition, Purpose, Scope, and Features
    - Assumptions and Dependencies
    - System Design and Implementation Approach and Data Model Specification
- Presentation 2 must cover at least the following aspects:
    - Project title, team member names, enrolment numbers, and guide name
    - Proposed System Definition, Purpose, Scope, and Features
    - Data Model
    - System Implementation
    - System Testing and Test case
    - Result Analysis

Presentation 1 evaluation parameters and its weightage are as follows:

Parameter	**Weightage %**
System Understanding	20
Identification of System Development Approach	20
Data Modelling	20
Work Sufficiency	10
Presentation skill (Technical)	15
Presentation skill (Linguistic)	05
Team Participation	10

Presentation 2 evaluation parameters and its weightage are as follows:

Parameter	**Weightage %**
System Implementation	30
System Evaluation	20
Implementation of suggestion given in Presentation 1	10
Presentation Skill (Technical and Linguistic)	20
Work Sufficiency	10
Team Participation	10

Annexure-I: Project Proposal Report Format

Shrimad Rajchandra Institute of Management and Computer Application

Uka Tarsadia University

MCA 3rd Semester

Course Name: Project **Course Code:**

Team Number					
Project Title					
Students guide interaction during	**Day**				**Total Hours per week**
	Time Duration				
Project Introduction					
Problem statements					
System Need/Objective					
System Functionalities including reports					
System Scope					
System limitation					
Tools and Technology					

Declaration:

We hereby declare that we have read the guidelines uploaded on the website for the above-said course and we shall follow those guidelines. We will not change the above-given details for the entire semester. I/We have chosen our team members by myself/ourselves. So, I/we will be responsible for team dispute, if any, during entire semester, which I/we will resolve by myself/ourselves.

We are well aware that project work must be prepared by the team members and must not be copied and (or) paid one. We acknowledge that the Institute has the authority for taking punitive actions against me for violation and/or non-compliance of the same.

Enrolment Number	Name	Signature

Team members were made aware by me regarding plagiarism and its consequences.

Guide Name	Signature

Date:

Place:

Annexure-II: Format of front Page

<<TITLE IN CAPITAL LETTERS>>

Submitted by,

<<Student's Name (Enrollment Number)>>,

<<Student's Name (Enrollment Number)>>

Guided by,

<<Guide Name>>

for partial fulfillment of the requirements

for the Degree of Master of Computer Application,

Shrimad Rajchandra Institute of Management and Computer Application,

Uka Tarsadia University,

<<Month>>, 2026.

Annexure III: Declaration Certificate

DECLARATION

We hereby declare that the project titled "<<Project Title>>" is fully implemented by us. It is neither paid nor copied. Even though, later on, in case of any infringement found for this project work, we are solely responsible for the same and understand that as per UGC norms, the University can revoke the degree conferred to us.

Enrolment No.	**Student Name**	**Signature**

As a guide, I assure you that there is no plagiarism found in the submitted document.

Guide Name:

Signature:

Date:

Place:

Annexure IV: Work Sufficiency Certificate

Work Sufficiency Certificate

As a guide, I assure you that the project work presented by the team is sufficient according to the specified time duration and team size.

Guide Name:

Signature:

Date:

Place: